

The Architecture of Assessment

Validating multiple-choice testing in an AI-enabled environment.



A structural framework for school administrators, teachers, and curriculum leaders.

The Diploma Reality

Social Studies 30-1 & 30-2

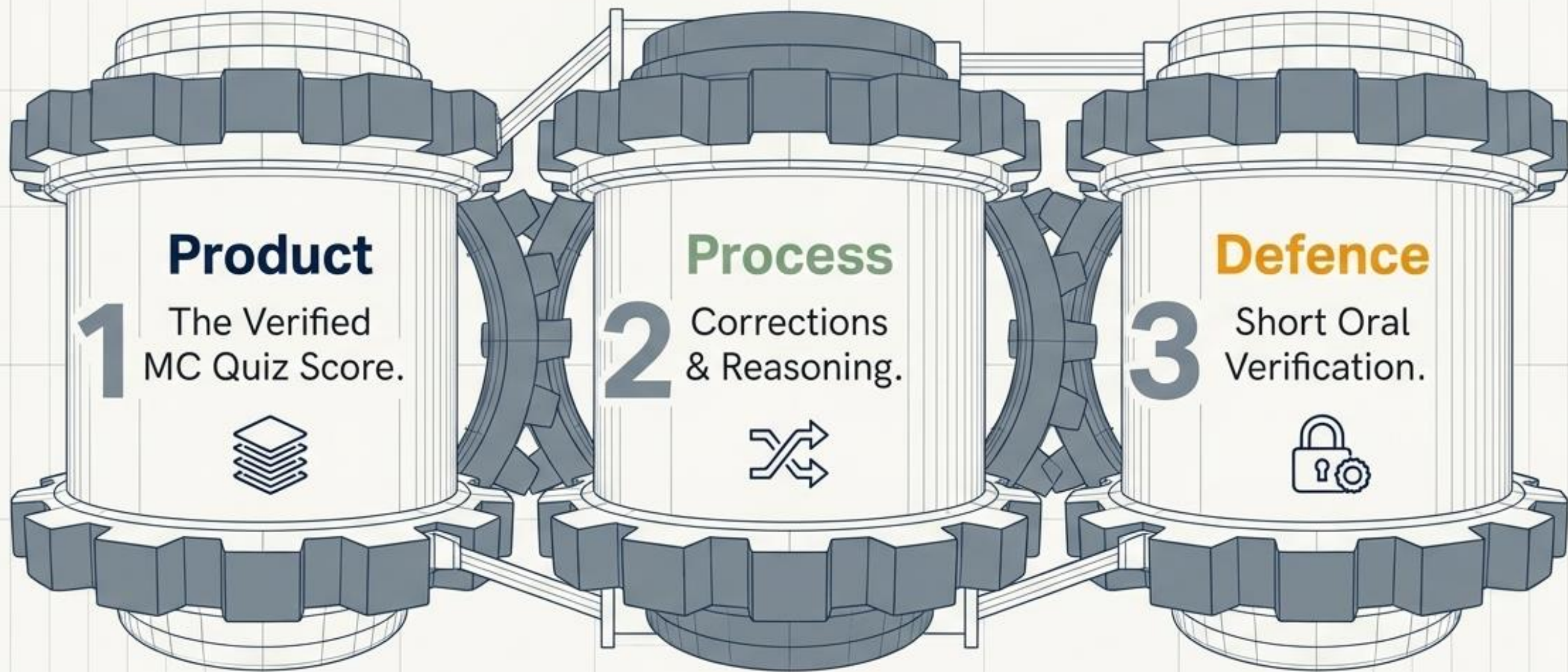
- **60** Multiple-Choice Questions
- **50%** of Diploma Exam Mark
- **30%** of Final Course Mark

The At-Home Reality

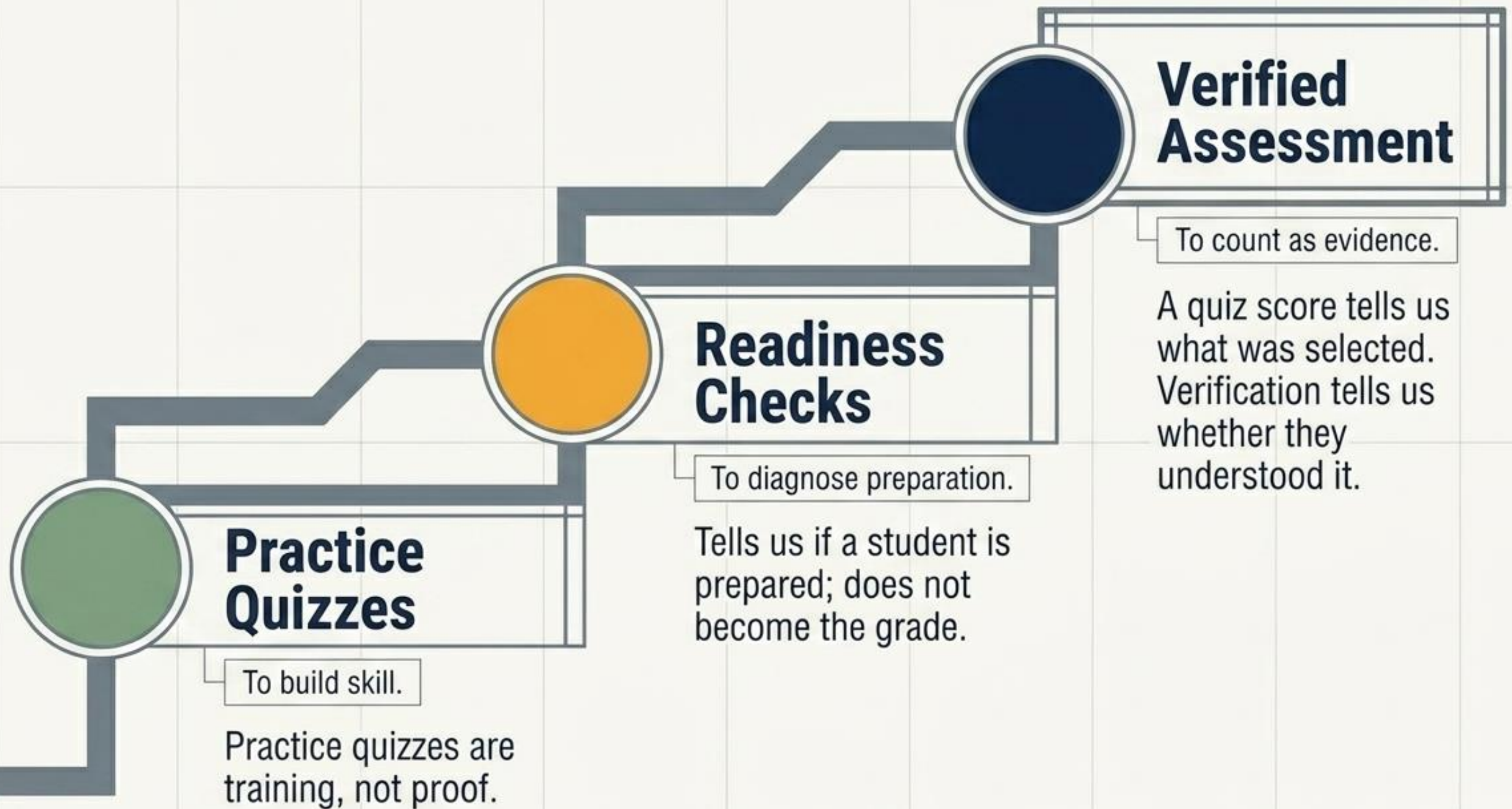


Unsupervised testing /
Instant AI-access /
Impossible to
definitively lock down.

Unsupervised at-home multiple choice
is not secure proof of learning by itself.
We cannot eliminate multiple choice,
so we must structurally validate it.

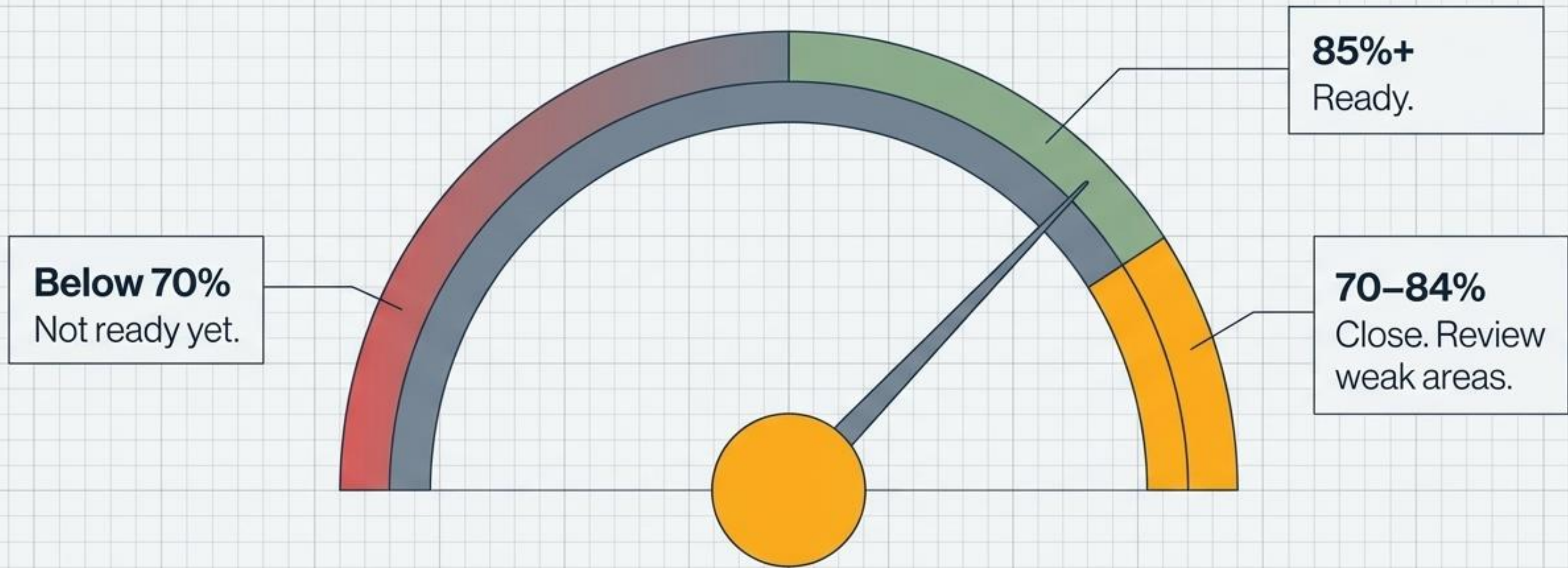


Multiple choice stays inside a robust assessment model instead of existing as a weak, standalone online quiz.



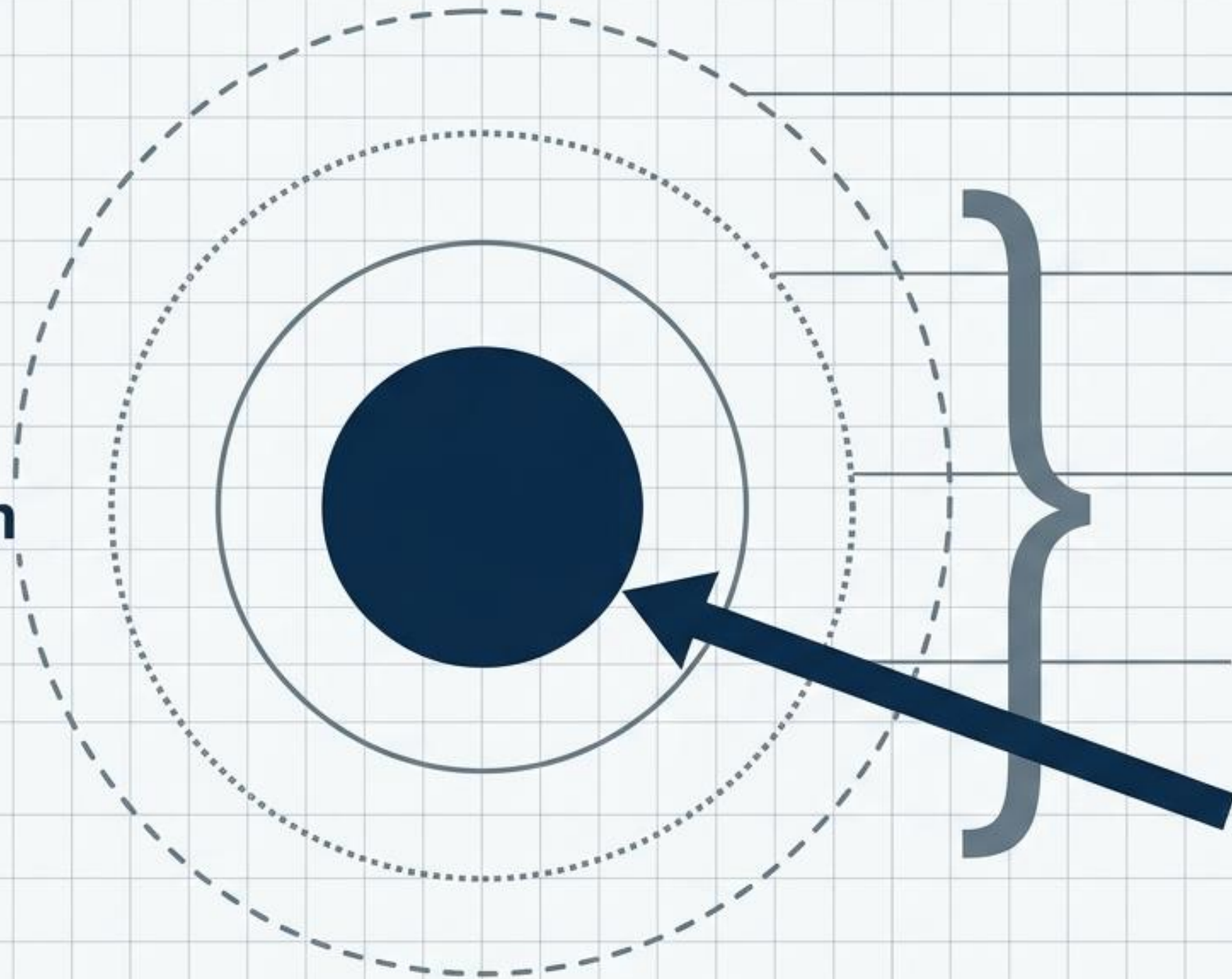
		Practice (Skill)	Readiness (Diagnosis)	Verified (Evidence)
Marks		No	No	Yes
AI Allowance		Allowed	Prefer No AI	Strictly Prohibited
Attempt Limits		Multiple	Usually One	One Attempt
Feedback Timing		Immediate	Score + General Feedback	No immediate answer reveal

 If students use AI during Practice, it does not damage assessment validity. They are either training honestly or wasting their own time.



Teaching self-monitoring. Readiness checks give teachers and students actionable data without pretending the score is secure evidence.

Friction



Layer 1 - Settings: Timed, randomized order/answers, large banks

Layer 2 - Design: Source-based questions, political cartoons, charts, specific Alberta/course contexts

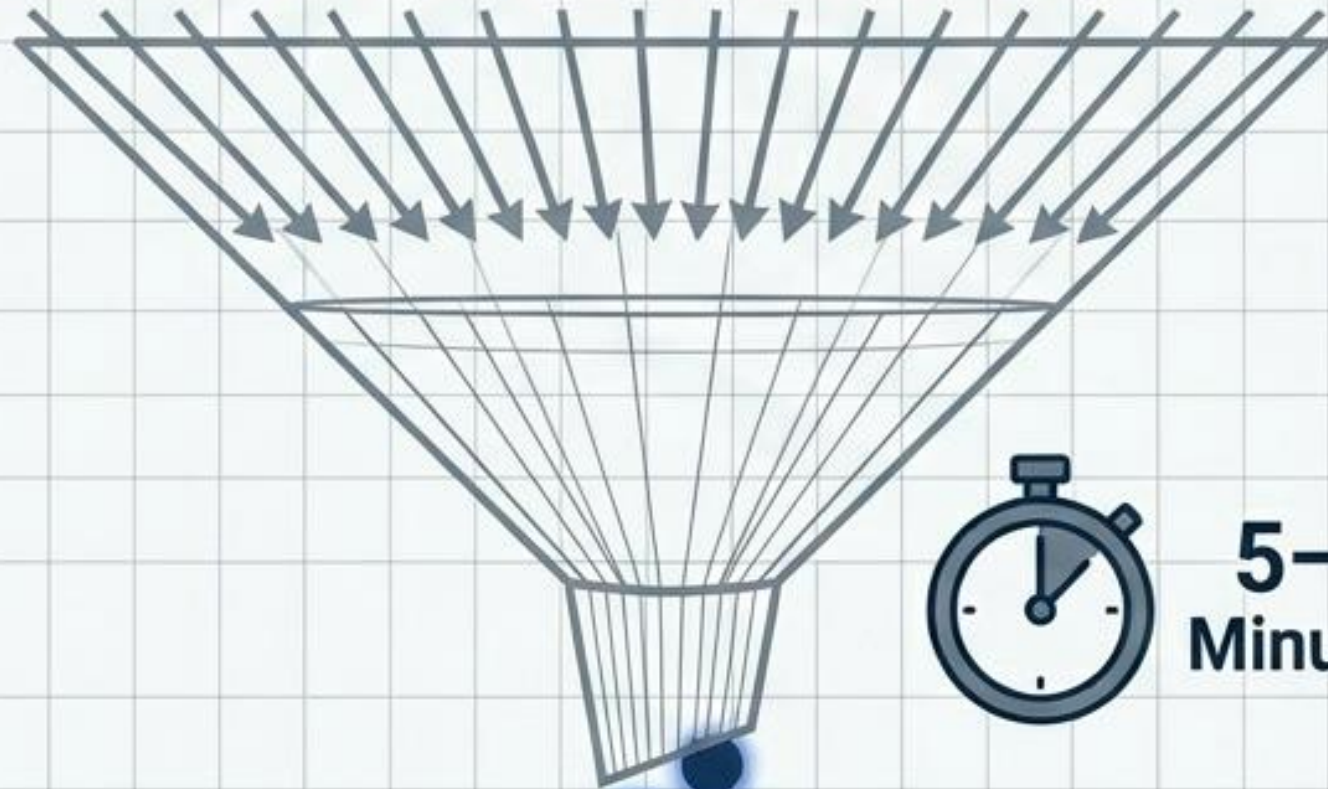
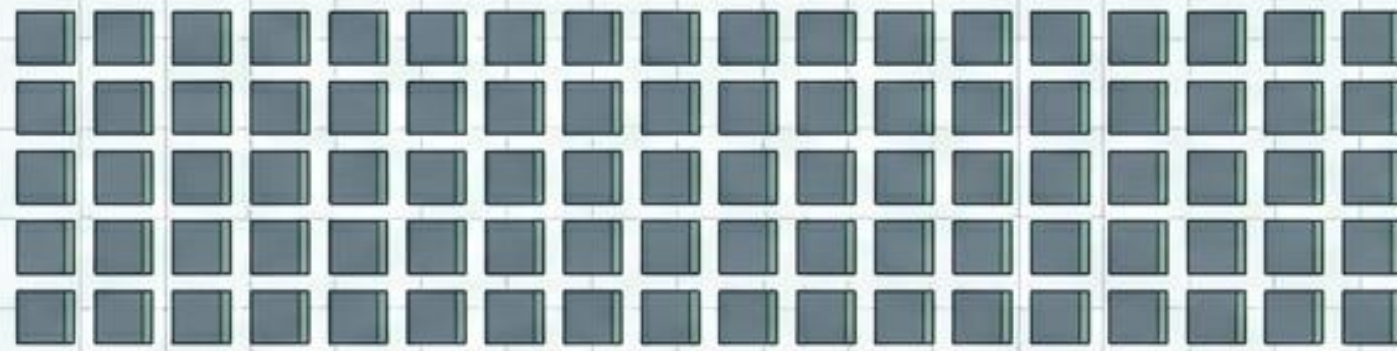
Layer 3 - Images: Using images for excerpts/questions to block simple copy-paste

Layer 4 - Verification: Oral Defence

Integrity

Images and settings add friction. They do not create security. Verification is the actual integrity layer.

60 Questions



**5-8
Minutes**

3-5 Targeted Questions

The Filter Output Breakdown:

- 1 question the student got correct
- 1 question the student got wrong
- 1 source-based question
- 1-2 important concept questions

Verification is not a retake. It is a highly targeted 5-minute sampling to prove authorship and comprehension.

The Defence Protocol: Making Understanding Visible

Defence Probe	COGNITIVE TARGET
Why did you choose that answer?	Checks: Reasoning
Why is one specific distractor wrong?	Checks: Concept understanding
What clue in the source mattered?	Checks: Source interpretation
What specific concept was being tested?	Checks: Course connection
Apply that idea to a new example.	Checks: Transfer

Student Explanation of Selected Answers

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graph TD; A[Student Explanation of Selected Answers] --> B[Explains well]; A --> C[Shaky but shows partial understanding]; A --> D[Clearly cannot explain answers]; B --> E[The Quiz Mark Stands.]; C --> F[Corrections Required (or reduce process portion)]; D --> G[Mark Capped or Reassessment Required.];
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Explains well

The Quiz Mark Stands.

Shaky but shows partial understanding

Corrections Required
(or reduce process portion)

Clearly cannot explain answers

Mark Capped or Reassessment Required.

The Absolute Rule: If the student cannot defend the result, the result is not fully validated.

Unified Institutional Messaging

School Policy

Practice and readiness checks are ungraded. Summative multiple-choice assessments count for marks only when paired with verification. Because online tests are completed at home, scores alone are not treated as complete proof of learning.

Student Expectation

You will get lots of practice to build diploma skills. Ungraded checks help you improve. When a test counts for marks, you must explain your answers. A high score without understanding will not hold up.

PRACTICE BUILDS SKILL.

READINESS CHECKS GUIDE PREPARATION.

VERIFIED M.C. PROVES UNDERSTANDING.

We are not trying to catch every possible cheat.
We are making understanding visible.